

Program : Diploma in Computer Application and Business Management	
Course Code : 3258	Course Title: Basic Statistical Analysis Lab
Semester : 3	Credits: 1.5
Course Category: Program Core	
Periods per week: 3 (L:0, T:0, P:3)	Periods per semester: 45

Course Objectives:

- To acquire knowledge on data analysis with the help of MS Excel.
- To provide students with the practical skills necessary for the analysis of economic and financial data.

Course Prerequisites:

Topic	Course code	Course name	Semester
Entire Course		Quantitative Techniques	3

Course Outcomes:

On completion of the course, students will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Apply data collection methods	6	Applying
CO2	Create Tables, Charts and Descriptive Statistics using Excel Functions	18	Applying
CO3	Implement the measures of central tendency & dispersion, and correlation using Analysis Toolpak in Excel	9	Applying
CO4	Implement linear regression using Analysis Toolpak in Excel	9	Applying
	Lab Exam	3	

CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3			3			
CO2	3			3			
CO3	3			3			
CO4	3			3			

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Apply data collection methods		
M1.01	List the stages of statistical enquiry in Descriptive Statistics	1	Remembering
M1.02	Perform primary data collection by preparing a questionnaire	2	Applying
M1.03	Perform secondary data collection from publications and historical documents	3	Applying
CO2	Create Tables, Charts and Descriptive Statistics using Excel Functions		
M2.01	Create frequency distribution table and histogram	3	Applying
M2.02	Summarize a table using PivotTable	3	Applying
M2.03	Perform questionnaire analysis using Excel	3	Applying
M2.04	Find the measures of central tendency and measures of dispersion a. Traditional formula b. Excel function	3	Applying
M2.05	Find the Karl Pearson Correlation coefficient using a. Traditional formula b. Excel function and display the scatter diagram	3	Applying
M2.06	Find the Spearman Rank Correlation coefficient using a. Traditional formula b. Excel function and display the scatter diagram	3	Applying
	Lab Exam – I	1 ½	

CO3	Implement the measures of central tendency & dispersion, and correlation using Analysis Toolpak in Excel		
M3.01	Install the Analysis Toolpak in Excel and use it to create frequency distribution table and histogram	3	Applying
M3.02	Find the measures of central tendency and measures of dispersion	3	Applying
M3.03	Find the Karl Pearson Correlation coefficient, Spearman Rank Correlation coefficient and display the scatter diagram	3	Applying
CO4	Implement linear regression using Analysis Toolpak in Excel		
M4.01	Perform linear regression and display the scatter diagram	3	Applying
M4.02	Evaluate the result of linear regression	3	Applying
M4.03	Open ended experiments	3	Applying
	Lab Exam – II	1 ½	

Text / Reference:

T/R	Book Title/Author
R1	Lisa M. Sullivan, Essentials of Biostatistics Workbook Statistical Computing Using Excel , Second Edition
R2	Dan Remenyi, George Onofrei, Joe English, An Introduction to Statistics Using Microsoft Excel
R3	T N Srivastava, Shailaja Rego , Statistics for Management
R4	David R. Anderson, Dennis J. Sweeney, Thomas A. Williams, Modern Business Statistics with Microsoft® Excel® ,
R5	Kiran Pandya and Smriti Bulsari, SPSS in simple steps , Dreamtech, 2011

Online Resources:

Sl.No	Website Link
1	http://www.textbooksfree.org/Statistics%20Book.htm
2	https://www.cengage.com/resource_uploads/downloads/0538733500_301118.pdf
3	https://www.wisdomjobs.com/e-university/excel-data-analysis-tutorial-1750.html
4	http://www.businessbookmall.com/Excel-Statistics-Book.htm
5	https://www.spss-tutorials.com/basics/